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1. Bluetooth

The BM96 is able to transfer stored measurement data via Bluetooth Low energy. It is also possible to read and change settings on the device. The BM96 always works as GAP peripheral / GATT server. Unless specified otherwise, the Implementation should be as specified from the Bluetooth SIG (<https://www.bluetooth.com/>).

Unless specified otherwise, all fields are in the order of LSO to MSO. Where LSO = Least Significant Octet and MSO = Most Significant Octet.

1.1 Security

The device uses Passkey Entry as a security mechanism. Therefore a 6 digit passkey has to be showed on the BM96 display, so the user can type it in the App. This value has to be generated randomly between 000 000 and 999 999. For further information please check the Bluetooth Core specification (4.0) Volume 3 Part H. It is important that the passkey entry is triggered before any important information (for example measurement data) can be read from the device or any settings can be changed. It should be possible to save 5 pairing information on the BM96, that means that it should be possible to connect up to 5 smartphones to the device and remember their pairing information. If a sixth smartphone is added the pairing information of the first smartphone should be replaced.

1.2 Advertising Package

The Advertising Package should contain information about the device type, name and its abilities.

The BM96 will start advertising either is a measurement has been made or if the users is entering the measurement menu on the device.

If no measurements are present the connection has to be established anyway, because the user could want to write some settings on the device. The advertisement packages should be sent for 30 seconds.

Name	Assigned Number	Description	Size
Flags	0x01	LE Limited discoverable mode and BR/EDR Not Supported: 0x05	
Incomplete List of 16-bit Service Class UUIDs	0x02	This List should show the Blood pressure Service: 0x1810	
Complete Local Name	0x09	The complete local name of the device "BM96": 0x42 4D 39 36	
Manufacturer Specific Data	0xFF	The first two byte represents the Company ID. In this case this should be 0x06 11 for Beurer. The next byte shows that the device is using the passkey feature and should be set to 0x01	
Appearance	0x19	The Appearance of the device should be Arm Blood Pressure meter (897)	

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1.3 Services and Characteristics

1.4 Services

These Services are used for the communication between the BM96 and the Smartphone Apps via Bluetooth.

Specification Name	Assigned Number	UUID	Note
Generic Access	0x1800	00001800-0000-1000-8000-00805f9b34fb	
Generic Attribute	0x1801	00001801-0000-1000-8000-00805f9b34fb	
Blood Pressure	0x1810	00001810-0000-1000-8000-00805f9b34fb	
Battery Service	0x2A19	00002A19-0000-1000-8000-00805f9b34fb	
Device Information	0x180A	0000180A-0000-1000-8000-00805f9b34fb	
Custom Service	0x2902	6e800001-b5a3-f393-e0a9-e50e24dcca9e	
Current Time Service	0x1805	00001805-0000-1000-8000-00805f9b34fb	as server

1.4.1 Service Characteristics

Generic Access

Name	Assigned Number	Properties	Note
Device Name	0x2A00	Read, Write	"BM96"
Appearance	0x2A01	Read	"897" –Blood Pressure: Arm
Peripheral Privacy Flag	0x2A02	Read, Write	Not supported
Reconnection Address	0x2A03	Read, Write	
Peripheral Preferred Connection Parameters	0x2A04	Read	Connection Interval: 7,5 - 30 ms Slave Latency: 0 Supervision Timeout Multiplier: 1000 (This is a suggestion and can be changed)

Generic Attribute

Name	Assigned Number	Properties	Note
Service Changed	0x2A05	Indicate	

Blood Pressure Service

Name	UUID	Properties	Note
Blood Pressure Measurement	0x2A35	Indicate	This characteristic has been changed and it not entirely BT SIG conform
Blood Pressure Feature	0x2A49	Read	

Device Information Service

Name	Assigned Number	Properties	Note
Manufacturer Name String	0x2A29	Read	"Beurer"
Model Number String	0x2A24	Read	"BM96"
Hardware Revision String	0x2A27	Read	Hardware Revision
Firmware Revision String	0x2A26	Read	Firmware Revision
Software Revision String	0x2A28	Read	Software Revision
System ID	0x2A23	Read	Manufacturer Identifier + Organizationally Unique Identifier
IEEE 11073-20601 Regulatory Certification Data List	0x2A2A	Read	Data
PnP ID	0x2A50	Read	Vendor ID Source Vendor ID Product ID Product Version

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Battery Service

Name	UUID	Properties	Note
Battery Level	0x2A19	Read	Level of Battery

Custom Service

Name	Assigned Number	Properties	Note
Record Control Point -Write	6e800002-b5a3-f393-e0a9-e50e24dcca9e	Write, Write no response	Used to send the Record Control Point commands to the BM96. It is possible to ask for the number of measurements, measurements and diagram data.
Record Control Point - Notify	6e800004-b5a3-f393-e0a9-e50e24dcca9e	Notify	Will send responses as notifications according to the Record Control Point commands
BT-MAC-Address	6e800003-b5a3-f393-e0a9-e50e24dcca9e	Read	Used to identify a device by the MAC address

1.5 Generic Access

For further information please visit <https://www.bluetooth.com/specifications/gatt/services> and check Generic Access (0x1800). Please Note the Name Field in Generic Access must contain the same name as the Advertising Package and the Device Information.

1.6 Generic Attribute

For further information please visit <https://www.bluetooth.com/specifications/gatt/services> and check the Generic Attribute (0x1801)

1.7 Blood Pressure

The BM96 is send the data after entering the measurement storage on the device or after a measurement has been made.

1.7.1 Blood Pressure Measurement

Blood Pressure Measurement – Indicate

For every measurement stored on the BM96 an indication is send. The BM96 should not end the Bluetooth connection after all measurements have been downloaded automatically.

Name	Format	Info	Note
flags	1 byte	See separate Flag table "Flag Description". Value is always 0xDE.	
systolic	sfloat (2byte)	To calculate the value, the bits must be split: The most significant 4 bit are the exponent and the other 12 bit the mantissa. The value is calculated by mantissa*10^(exponent). With the BM96 the exponent is always 0xF-> -1. This means the exponent would be always 10^(-1). The value is in mmHg. The valid range is 0-300.	
diastolic	sfloat (2byte)	To calculate the value, the bits must be split: The most significant 4 bit are the exponent and the other 12 bit the mantissa. The value is calculated by mantissa*10^(exponent). With the BM96 the exponent is always 0xF-> -1. This means the exponent would be always 10^(-1). The value is in mmHg. The valid range is 0-300.	
mean arterial pressure	sfloat (2byte)	Not supported by the BM96. The value has always to be 0x07FF	
timestamp	time (7byte)	See table "Timestamp description" for the timestamp data	
pulse rate	sfloat (2byte)	To calculate the value, the bits must be split: The most significant 4 bit are the exponent and the other 12 bit the mantissa. The value is calculated by mantissa*10^(exponent). With the BM96 the exponent is always 0xF-> -1. This means the exponent would be always 10^(-1). The value is in bpm.	
User Id	1 byte	The Id of the user. 0x00 means user 1 0x01 means user 2	

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measurement status	2 byte	For more information, please check the table "Measurement Status" below.	

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Flag description – Part of Blood Pressure Measurement

The flags have a size of 1 byte and are part of the blood pressure measurement Indication. The Flags of the BM96 are 0xDE

Name	Format	Info	Note
unit flag	1 bit	systolic, diastolic and MAP unit: 0 = mmHg 1 = kPa	
time stamp flag	1 bit	0 = not present 1 = present	
pulse rate flag	1 bit	0 = not present 1 = present	
user ID flag	1 bit	0 = not present 1 = present	
measurement status flag	1 bit	0 = not present 1 = present	
reserved for future use	1 bit		
HSD flag	1 bit	0 = not present 1 = present	
AFIB flag	1 bit	0 = not present 1 = present	

Measurement Status – Part of Blood Pressure Measurement

The measurement status has a size of 2 bytes and is part of the blood pressure measurement Indication.

Name	Format	Info	Note
body movement detection	1 bit	0 = no body movement 1 = body movement	Feature not supported!
cuff fit detection	1 bit	0 = cuff fits properly 1 = cuff is too loose	Feature not supported!
irregular pulse	1 bit	0 = no irregular pulse 1 = irregular pulse	
pulse rate range detection	2 bit	00 = in range 01 = exceeds upper limit 10 = below lower limit 11 = reserved for future use	Feature not supported!
measurement position detection	1 bit	0 = proper position 1 = improper position	Feature not supported!
HSD	2 bit	10 = HSD not detected 11 = HSD detected 01 = unable to detect	The Feature is not defined by Bluetooth SIG.
AFIB	1 bit	1 = AFIB 0 = no AFIB	The Feature is not defined by Bluetooth SIG.

Timestamp description – Part of Blood Pressure Measurement

The timestamp has a size of 7 bytes and is part of the blood pressure measurement Indication.

Name	Format	Info	Note
year	uint16 (2 byte)	The value of 1 equals 1 year. The valid range is from 1582-9999	
month	uint8 (1 byte)	The valid range is from 0-12. The value 0 represents unknown, the value of 1 represents January, the value of 2 represents February ...	
day	uint8 (1 byte)	The valid range is from 0-31. The value 0 represents unknown, the value of 1 represents the first day, the value of 2 represents the second day...	
hours	uint8 (1 byte)	The value of 1 equals 1 hour. The valid range is from 0-23	
minutes	uint8 (1 byte)	The value of 1 equals 1 minute. The valid range is from 0-59	
seconds	uint8 (1 byte)	The value of 1 equals 1 second. The valid range is from 0-59	

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1.7.2 Blood Pressure Feature

Blood Pressure Feature – Read

The blood pressure feature tells the user which features are supported. The value for the BM96 is always [0xC4 00]. That means the BM96 supports the irregular pulse feature and HSD.

Name	Format	Info	Note
body movement feature	1 bit	0 = feature not supported 1 = feature supported	
cuff fit feature	1 bit	0 = feature not supported 1 = feature supported	
irregular pulse feature	1 bit	0 = feature not supported 1 = feature supported	
pulse rate range	1 bit	0 = feature not supported 1 = feature supported	
measurement position	1 bit	0 = feature not supported 1 = feature supported	
multiple bond feature	1 bit	0 = feature not supported 1 = feature supported	
HSD feature	1 bit	0 = feature not supported 1 = feature supported	
AFIB feature	1 bit	0 = feature not supported 1 = feature supported	
reserved for future use	8 bit		

1.7.3 Battery Service

The Battery Level can be asked by the App with a Read request. Additionally the Battery level should be send with a Notification as soon as a Battery warning is shown on the BM96. So the App can show the warning at the same time. For further information please visit <https://www.bluetooth.com/specifications/gatt/services> and check the Battery Service (0x180F)

1.8 Custom Service

The Custom Service is used for different device options that are not provided by the Bluetooth standard profiles. It is used to get the ECG data from the BM96

Record Control Point – Write

Use this Record Control Point to interact with the measurement record data on the device.

Name	Format	Info	Note
OP code	1 byte	0x01 – Record overview 0x02 – Get Measurements 0x03 – Get Diagram data	
Operator	2 byte	Operator for OP-Code 0x01: No operator necessary, always send 0x00 00 Operator for OP-Code 0x02: 0x00 00: Get all measurements of User 1 0x10 00: Get all measurements of User 2 0x01 00: Get new measurements (BT) of User 1 0x11 00: Get new measurements (BT) of User 2 0xFF XX: Confirm the measurement. XX should be set to the received measurement ID. After the confirmation has been send the BM96 will change the new measurement flag byte to 0. Operator for OP-Code 0x03: 0xYY ZZ: The first byte YY should be set to the measurement ID of the measurement from which the Diagram data is required. In the second byte it should be possible to configure the amount of data point. 0x YY 01: All diagram data are sent 0x YY 02: The mean value of two diagram points is calculated and the mean value are transferred 0x YY 03: The mean value of three diagram points is calculated and the mean values are transferred 0x YY 04: The mean value of four diagram points is calculated and the mean values are transferred	

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Record Control Point – Notify

If the user has send **OP Code 0x01** (Daily record presence) to the Record Control Point the following Notification will be received:

Name	Format	Info	Note
Header	1 byte	0x01	
Amount of overall ECG-Measurements User1	1 byte	The amount of overall ECG measurements of user 1	
Amount of overall ECG-Measurements User2	1 byte	The amount of overall ECG measurements of user 2	
ECG-Measurements with BT new flag User1	1 byte	The amount of ECG measurements that are marked as new BT measurement of user 1	
ECG-Measurements with BT new flag User2	1 byte	The amount of ECG measurements that are marked as new BT measurement of user 2	

If the user has send **OP Code 0x02** (Get Measurements) to the Record Control Point, depending on the Operator the latest measurement of the new measurements or all measurements is transferred.

Name	Format	Info	Note
Header	1 byte	0x02	
Measurement ID	1 byte	A unique ID that is used to refer to the specific measurement with other commands. The ID is only valid during one connection. It is possible that another measurement gets the same ID if the first one has been deleted. The most significant bit will be 0 if the measurement belongs to user 1. If it belongs to user 2 the most significant bit will be 1.	
New Measurement Flag	1 byte	If the measurement is a new one (not downloaded) via BT the lowest four bit are 1 (0x XF) otherwise they are 0 (0x X0). If the measurement is a new one (not downloaded) via USB the highest 4 bit are 1 (0x FX) otherwise they are 0 (0x 0X)	
Measurement duration	1 byte	The duration of the measurement in seconds	
Timestamp	7 byte	The time Stamp is the same one that has been used for the blood pressure. Please check the time stamp description there for more details.	
Status Flags	1 byte	The BM96 can show different status flags. If the bit is 1 means that the status is present 0 means it is not present: Bit 0: OK Bit 1: Fast Bit 2: Pause (not implemented in BM96) Bit 3: Slow Bit 4: Wave (not implemented in BM96) Bit 5: Rhythm (not implemented in BM96) Bit 6: PVC (not implemented in BM96) Bit 7: AFIB	
RR-Interval	2 byte		
Puls	2 byte	The Puls value in beats per minute	

In case there is no data present the following notification is sent:

Name	Format	Info	Note
Header	1 byte	0x02	
Index	1 byte	0xFF	

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If the user has send **OP Code 0x03** (Get Diagram data) to the Record Control Point, the device will respond with the diagram data to the according measurement. There will be two different kinds of notifications. The first one contains some general information and the first ECG data:

Name	Format	Info	Note
Header/Index	1 byte	0x03 The Header in this case consists of two parts. The 4 lower bits will the Header and show that this is a Diagram point notification. They will always be 3 for Diagram point notifications. The upper 4 bit will be an index that counts up the notifications. So this first notification will be 0x03, the next one 0x13, the next 0x23 etc. . After 16 Notifications it will start with 0x03 again. This value has to be checked to make sure the notifications are in the correct order.	
Measurement ID	1 byte	The ID of the measurement that the diagram data belongs to	
Diagram points	1 byte	The amount of points that have been transferred 0x01 means every point has been transferred 0x02 means every second point had been transferred ... 0x0A means every tenth point has been transferred	
Time interval	1 byte	The time difference between the ECG values in milliseconds after the mean value has been calculated.	
Length	4 byte	Number of ECG data points	
First ECG Value	2 byte	The value is signed. It can be calculated like a short in Java: 0x00 00 to 0x7F FF represent the positive values from 0 to 32767. The value 0xFF FF is the highest negative number -1 and 0x80 00 would represent the lowest number -32768. To get the ECG value it has to be multiplied by 10 µV.	
Second ECG Value	2 byte		
Third ECG Value	2 byte		
...	...	...	
Sixth ECG Value	2 byte		

The second kind of notification will be sent multiple times until all ECG Diagram points are received:

Name	Format	Info	Note
Header/Index	1 byte	0x03 The Header in this case consists of two parts. The 4 lower bits will the Header and show that this is a Diagram point notification. They will always be 3 for Diagram point notifications. The upper 4 bit will be an index that counts up the notifications. So this first notification will be 0x03, the next one 0x13, the next 0x23 etc. . After 16 Notifications it will start with 0x03 again. This value has to be checked to make sure the notifications are in the correct order.	
Measurement ID	1 byte	The ID of the measurement that the diagram data belongs to	
ECG Value	2 byte		
ECG Value	2 byte		
ECG Value	2 byte		
...	...	...	

In case there is no data present the following notification is sent:

Name	Format	Info	Note
Header	1 byte	0x03	
Day/Notification index	1 byte	0xFF	

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Unfortunately, we found some issue regarding the ECG download that can occur in devices with Firmware Version 44 or lower:

If there are more than 60 Blood pressure measurements (User 1 and 2 combined) there is some storage overflow, and the download doesn't work anymore. The device will show Err6 on the display and disconnect. If this happens it is still possible to connect to the device, but the download of measurements won't work.

Recommended workaround:

We recommend deleting the measurements after the download. If someone is downloading the values regularly this should prevent the issue. Since this bug only influences the download and not the stored measurements, it won't be an issue for users that don't use the app at all.

For the users that are using the App very infrequently and run into this issue, we recommend showing some message in the app in case the BM96 does disconnect when trying to download measurement data. In this Dialog the user should check if the device shows Err6 when it disconnects and if yes, the device should be reset manually by the user as described in the user manual.

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Delete Measurements – Write

Name	Format	Info	Note
User ID	1 byte	0x00 to delete Measurements of User 1 and 2 0x01 to delete Measurements of User 1 0x02 to delete Measurements of User 2	
Measurements	1 byte	0x00 Delete ECG and blood pressure measurements 0x01 Delete only ECG data. Please do not use this command in FW version <= V44 0x02 Delete only blood pressure measurements Please do not use this command in FW version <= V44	

We found some issue with reindexing the remaining measurements that are still on the device, if only ECG or blood pressure measurements are deleted. This is fixed in Firmware version V45. Please don't use this command in a lower FW version. In case measurements must be deleted, please delete all measurements.

Delete Measurements – Notify

In case of **OP Code 0x04** (Delete stored records) the user will get a success or fail message for the delete command.

Name	Format	Info	Note
User ID	1 byte	0x00 to delete Measurements of User 1 and 2 0x01 to delete Measurements of User 1 0x02 to delete Measurements of User 2	
Measurements	1 byte	0x00 Delete ECG and blood pressure measurements 0x01 Delete only ECG data 0x02 Delete only blood pressure measurements	
Confirmation	1 byte	0x00 Delete Measurement was successful 0x01 No measurements found to delete 0x02 Delete unsuccessful	

After the measurements have been deleted, it is still possible to see them in the measurement storage until the BM96 has been shut down.

BT-MAC-Address – Read

Name	Format	Info	Note
MAC-Address	6 byte	The 6 byte Bluetooth MAC Address of the device.	

The MAC-Address is used to identify a specific device by the App.

1.9 Current Time Service

The current time Service is used to set the current time from the smartphone to the device. If no time is set the device is using a default time that is not running. The default time should be set in case the scale has lost the time and has not been synchronized with a smartphone (Battery change, new device, reset etc) so the app knows that the measurements don't have a valid timestamp.

Name	Format	Info	Note
Year	2 byte	The current year. 0 means the year is unknown	
Month	1 byte	The current month. Valid values are 0-12: 0 = unknown/default value 1 = January 2 = February 3 = March 4 = April 5 = Mai 6 = June 7 = July 8 = August 9 = September 10 = October 11 = November 12 = December	
Day	1 byte	The current day of the month. 0 means the day is unknown and is used as default value. Valid values are 0-31.	
Hours	1 byte	The current hour. Valid values are 0-23. Use the value 0 as default.	
Minutes	1 byte	The current minute. Valid values are 0-59. Use the value 0 as default.	
Seconds	1 byte	The current second. Valid values are 0-59. Use the value 0 as default.	

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		0 = unknown /default value 1 = Monday 2 = Tuesday 3 = Wednesday 4 = Thursday 5 = Friday 6 = Saturday 7 = Sunday	
Day of week	1 byte		
Fraction 256	1 byte	Not used. Set to 0x00	
Adjust Reason	1 byte	Bit 0: 0 = No manual time update 1= Manual time update Bit 1: 0 = No external time update 1= External time update Bit 2: 0 = No change of time zone 1= Change of time zone Bit 3: 0 = No DST(Daylight Saving Time) change 1 = DST change Bit 4,5,6,7 set to 0	

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2. USB

2.1 General Information

The blood pressure monitor BM96 is able to transfer stored measurement data via USB. The device is a HID device, so no driver installation is necessary. The Vendor ID (VID) is: 0x314A and the Product ID (PID) is: 0x0102. The device appears as an USB 2.0 low speed device with interrupt transmit. The data length is 8 bytes for OUT report transfer and 64 byte for IN report transfer.

If the data length is less than the specified bytes the data is filled with 0xF4.

OUT report transfer is managed over Endpoint 0 Control Transfer

IN report transfer is manager over Enpoint 1 Interrupt Transfer with 10 ms update rate. This is necessary for fast diagram transfer.

It is possible that the device does not respond immediately after a command has been send.

2.2 Connection

This command starts a connection between the PC and the BM96.

Name	Format	Info	Note
Command	1 byte	The command byte is 0xAA	
Filler bytes	7 byte	The following bytes are filler bytes with 0xF4	

The response from the device to the connection attempt.

Name	Format	Info	Note
Response	1 byte	The response for the connection attempt is 0x55	
Filler bytes	63 byte	The following bytes are filler bytes with 0xF4	

2.3 Get Name of device

This command asks for the name of the device. The name of the device is "Blood Pressure Monitor BM96". The Name is transmitted in 4 parts. Each part has 8 bytes and is triggered with another command. You have to send all 4 commands to receive the complete name

Name	Format	Info	Note
Command	1 byte	The Command byte of the name is: 0xA1	
Filler bytes	7 byte	The following bytes are filler bytes with 0xF4	

This is the response of the device. Depending on which command has been send you will receive a part of the device name. Every part is 8 byte long, the last part doesn't have 8 characters and is filled with 0xF4.

Name	Format	Info	Note
Response	34 byte	ASCII encoded: "Beurer Blood Pressure Monitor BM96"	
Filler bytes	28 byte	The following bytes are filler bytes with 0xF4	

2.4 Number of measurements

This command request for the number of measurements from the device.

Name	Format	Info	Note
Command	1 byte	The command to request the amount of measurements is 0xA2	
Filler bytes	7 byte	The following bytes are filler bytes with 0xF4	

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The response from the device.

Name	Format	Info	Note
Measurement amount user 1	1 byte	The amount of all measurements on the device for user 1 [0 ... 60]	
Measurement amount user 2	1 byte	The amount of all measurements on the device for user 2 [0 ... 60]	
Filler bytes	62 byte	The following bytes are filler bytes with 0xF4	

2.5 Read a measurement diagnosis

This command is used to read a measurement on the device.

Name	Format	Info	Note
Command	1 byte	The command to read a measurement is 0xA3	
User id	1 byte	The user ID. 0 = user 1 and 1 = user 2	
Measurement ID	1 byte	the measurement ID must be less than the value received from number of measurements	
Filler bytes	5 byte	The following bytes are filler bytes with 0xF4	

The response from the device in case of blood pressure measurement:

Name	Format	Info	Note
user_id	1 byte	The user ID. 0 = user 1 and 1 = user 2	0
Measurement ID	1 byte	the measurement ID of the requested measurement	1
New Measurement Flag	1 byte	If the measurement is a new one (not downloaded) via BT the lowest four bit are 1 (0x XF) otherwise they are 0 (0x X0). If the measurement is a new one (not downloaded) via USB the highest 4 bit are 1 (0x FX) otherwise they are 0 (0x 0X)	2
measurement type	1 byte	1 = BP	3
hour	1 byte	the hour the measurement was taken	4
minute	1 byte	the minute the measurement was taken	5
day	1 byte	the day the measurement was taken	6
month	1 byte	the month the measurement was taken	7
year	1 byte	the year the measurement was taken	8
Flags	1 byte	Bit 0,1	10 = HSD not detected 11 =HSD detected 01 = unable to detect IRH 1=True 0=False AFIB 1=True 0=False
		Bit 2	
		Bit 3	
Systolic	2 byte	systolic pressure in mmHg MSB first	10,11
Diastolic	1 byte	diastolic pressure in mmHg	12
Pulse	1 byte	heartrate in bpm	13
Filler bytes	all other bytes	The following bytes are filler bytes with 0xF4	

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The response from the device in case of ecg measurement:

Name	Format	Info	Note
user_id	1 byte	The user ID. 0 = user 1 and 1 = user 2	
Measurement ID	1 byte	the measurement ID must be less than the value received from number of blood pressure measurements	
New Measurement Flag	1 byte	If the measurement is a new one (not downloaded) via BT the lowest four bit are 1 (0x XF) otherwise they are 0 (0x X0). If the measurement is a new one (not downloaded) via USB the highest 4 bit are 1 (0x FX) otherwise they are 0 (0x 0X)	
measurement type	1 byte	2 = ECG	
hour	1 byte	the hour the measurement was taken	
minute	1 byte	the minute the measurement was taken	
day	1 byte	the day the measurement was taken	
month	1 byte	the month the measurement was taken	
year	1 byte	the year the measurement was taken	
Flags	1 byte	Bit7 = WAVE Bit6 = AFIB Bit5 = PVC Bit4 = PAUSE Bit3 = RHYTHM Bit2 = OK Bit1 = FAST Bit0 = SLOW	
heart rate	2 byte	heartrate in bpm. MSB first	
Filler bytes	all other bytes	The following bytes are filler bytes with 0xF4	

2.6 Read ECG diagram package count

This command is used to read the number of packages that need to be transferred for this specific ecg diagram. A diagram package is described in chapter 2.7

Name	Format	Info	Note
Command	1 byte	The command to read a measurement is 0xA4	
User ID	1 byte	The user ID. 0 = user 1 and 1 = user 2	
Measurement ID	1 byte	The measurement ID must be less than the value received from number of measurements	
Filler bytes	6 byte	The following bytes are filler bytes with 0xF4	

The response from the device.

Name	Format	Info	Note
User ID	1 byte	The user ID. 0 = user 1 and 1 = user 2	
Measurement ID	1 byte	The measurement ID of the requested measurement	
Diagram package amount	2 byte	Number of diagram packages. MSB first	
Filler bytes	60 byte	The following bytes are filler bytes with 0xF4	

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2.7 Read ECG diagram

This command is used to read a specific diagram package.

Name	Format	Info	Note
Command	1 byte	The command to read a measurement is 0xA5	
User ID	1 byte	The user ID. 0 = user 1 and 1 = user 2	
Measurement ID	1 byte	The measurement ID must be less than the value received from number of measurements	
Package index	2 byte	A number between 0 and the diagram package count. LSB first	
Filler bytes	3 byte	The following bytes are filler bytes with 0xF4	

The response from the device.

Name	Format	Info	Note
Start Byte	1 byte	0xA5	
Measurement ID	1 byte	The measurement ID must be less than the value received from number of measurements	
Package ID	2 byte	A number between 0 and the diagram package count. LSB first	
1. Data point	2 byte	Data point in 10 uV. LSB first	
...	...	...	
30. Data point	2 byte	Data point in 10 uV. LSB first	

2.8 Delete measurements

If you want to delete all measurements for a specific user you can use command 0xA6

Name	Format	Info	Note
Command	1 byte	The command to delete is 0xA6	
User ID	1 byte	The user ID. 0 = user 1 and 1 = user 2	
Filler bytes	6 byte	The following bytes are filler bytes with 0xF4	

The response from the device.

Name	Format	Info	Note
Command	1 byte	0x55 = delete success	
Filler bytes	63 byte	The following bytes are filler bytes with 0xF4	

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2.9 Validate measurement

To change the New Measurement Flag for USB send 0xA7 command. If the measurement is a new one (not downloaded) via BT the lowest four bit are 1 (0x XF) otherwise they are 0 (0x X0). If the measurement is a new one (not downloaded) via USB the highest 4 bit are 1 (0x FX) otherwise they are 0 (0x 0X)

Name	Format	Info	Note
Command	1 byte	The command to delete is 0xA7	
User ID	1 byte	The user ID. 0 = user 1 and 1 = user 2	
Measurement ID	1 byte	0-59	
Filler bytes	6 byte	The following bytes are filler bytes with 0xF4	

The response from the device.

Name	Format	Info	Note
Command	1 byte	0x55 = validation success	
Filler bytes	63 byte	The following bytes are filler bytes with 0xF4	

2.10 Turn of current communication

If you want to end the communication between the device and BM96 send 0xF7 command.

Name	Format	Info	Note
Command	1 byte	The command to turn of the communication is 0xF7	
Filler bytes	7 byte	The following bytes are filler bytes with 0xF4	

HISTORY	Changes to the previous document will be marked in red and documented below. Before updating the document, all previous red markings have to be set back to black colour.		Issued/ changed by
Revision	Date	Description	
01	2019-08-05	Initial version	ULAL
02	2020-03-27	only comments added	ULAL
03	2020-04-23	new USB protocol	ULAL
04	2020-05-29	added userid to USB package 2.7	ULAL
05	2020-06-05	changed to 10 ms update rate	ULAL
06	2020-07-15	added current Time service, added AFIB in Blood Pressure, added UUID of custom service	ULAL
07	2020-08-21	fixed mistake in blood pressure feature	ULAL
08	2020-09-25	added new measurement flag for usb	ULAL
09	2021-02-16	added HSD_NA in USB	ULAL
10	2021-11-04	Changed Flag value of BPM added explanation for calculating sfloat value	KEMA
11	2021-11-15	Changed Characteristic overview and added UUIDs	KEMA
12	2022-05-05	Changed USB name	KEMA
13	2022-05-19	Corrected Service UUID + Explanation of Diagram point calculation	KEMA
14	2022-09-19	Added info which ECG Flags are supported, added issue description and information for delete command, corrected byte order for some USB values, added missing "new measurement" flag in USB, download issue + workaround described.	KEMA
15	2022-09-23	Byte order corrected to LSO first on USB ECG Read command	KEMA